

LLM Training Pipeline and Deployment Design

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1. Design Objectives

- Ensure responses are grounded in validated and peer-reviewed medical sources with visible citations.
- Maintain educational-only responses without diagnosis or treatment recommendations.
- Reduce hallucinations and bias using Retrieval-Augmented Generation (RAG), guardrails, and dataset filtering.
- Meet performance requirements such as response time under 30 seconds and system uptime above 95%.

2. Training Pipeline Overview

- Data ingestion and governance for collecting and validating trusted medical datasets.
- Dataset preparation separating knowledge sources and behavioral training examples.
- Optional supervised fine-tuning focused on safety, refusal behavior, and response formatting.
- Evaluation pipelines to measure groundedness, bias risk, hallucination rates, and latency.

3. Data Ingestion and Governance

- Register data sources and maintain metadata including origin, license, and reliability.
- Apply de-identification and privacy checks to prevent storing sensitive information.
- Filter low-quality or non-cited conversational data that may increase hallucination risk.
- Tag datasets for known bias risks such as geographic or demographic imbalance.

4. Dataset Preparation

- Knowledge Track: convert trusted sources into text chunks and generate embeddings for vector search.
- Store document metadata such as title, authors, year, and publication source.
- Behavior Track: create examples for refusal responses, disclaimers, and citation formatting.
- Train the model to ask clarifying questions when prompts are ambiguous.

5. Model Training Strategy

- Use a pretrained large language model as the base model.
- Apply light supervised fine-tuning to reinforce safe and consistent response behavior.
- Use RAG as the primary mechanism for factual medical knowledge retrieval.
- Require the model to ground responses in retrieved passages before generating answers.

6. Evaluation and Validation

- Test groundedness by verifying responses match retrieved knowledge sources.
- Evaluate hallucination risk using prompts designed to test unsupported claims.
- Assess bias by testing responses across different populations or contexts.
- Measure performance against system latency and accuracy thresholds before release.

7. Deployment Architecture

- Frontend web application providing the chat interface and avatar visualization.
- Backend API services handling chat requests, guardrails, retrieval, generation, and feedback.

- Retrieval service using embeddings and vector search for knowledge retrieval.
- Text-to-speech service for optional audio output.

8. Data Storage Components

- Vector database for embeddings and semantic search.
- Document storage for validated medical source content.
- Feedback database for user interaction ratings and comments.
- System logs and metrics for monitoring performance and reliability.

9. Runtime Request Flow

- User submits a prompt through the web interface.
- The backend processes the request and checks guardrails for scope compliance.
- The retrieval service finds relevant medical knowledge sources.
- The LLM generates an answer grounded in retrieved information.
- Citations are appended to the response and returned to the user.
- Optional audio output is generated through the text-to-speech service.
- User feedback is collected for continuous improvement.

10. Monitoring and Continuous Improvement

- Monitor hallucination rates, refusal accuracy, and citation coverage.
- Track user feedback to identify improvement opportunities.
- Update the retrieval knowledge base with new peer-reviewed sources.
- Run evaluation tests before deploying updated models.